

- 5 The I - V characteristic of an electrical component X is shown in Figure 5.1.

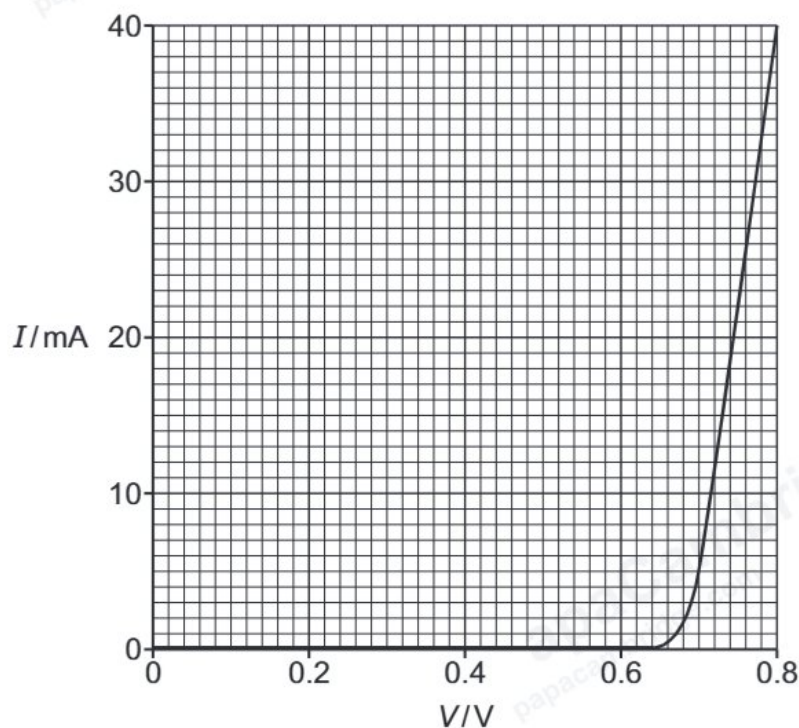


Figure 5.1

- (a) State the name of electrical component X.

..... [1]

- (b) A metallic conductor has a constant resistance of $40\ \Omega$. On Figure 5.1, sketch the I - V characteristic of this conductor between $V = 0$ and $V = 0.80\text{ V}$.

[2]

- (c) Component X is connected in a circuit with a resistor of resistance $20\ \Omega$ as shown in Figure 5.2.

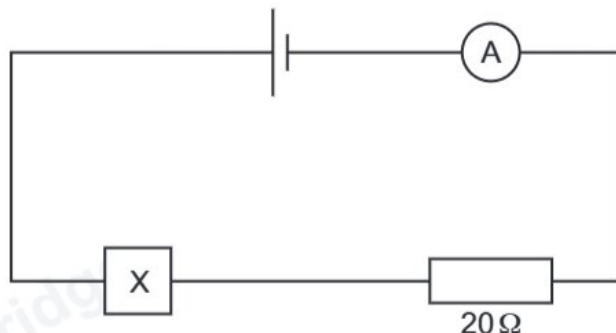


Figure 5.2

The current in the ammeter is $5.0\ \text{mA}$ and the cell has negligible internal resistance.

Show that the electromotive force (e.m.f.) of the cell is $0.80\ \text{V}$.

[3]

- (d) The terminals of the cell in **5(c)** are now reversed without making any other changes to the circuit.

State the reading on the ammeter.

reading on ammeter =A [1]

- (e) The resistor of resistance $20\ \Omega$ and component X are now connected in parallel to the cell in 5(c) in the circuit shown in Figure 5.3.

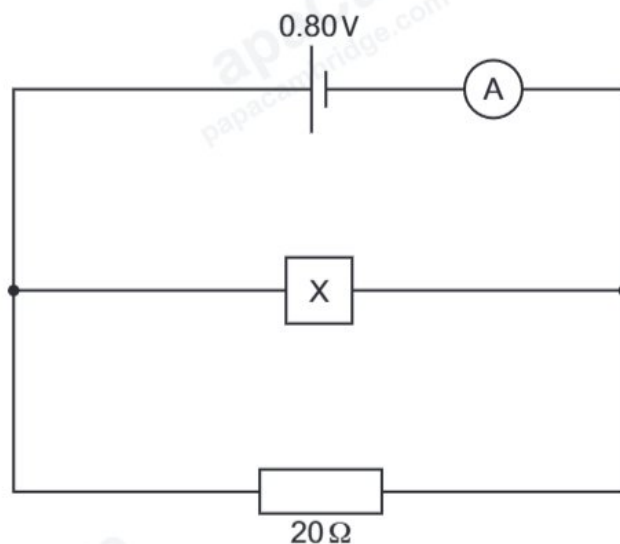


Figure 5.3

The current in component X is **not** negligible.

Use Figure 5.1 to determine the current in the ammeter.

current =A [3]

[Total: 10]